

Advanced Machine Learning

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Assistant Professor

Department of Computer Engineering

Faculty Profile – Nirmala Balookar

□ Educational Qualifications

- Pursuing Ph.D. in Computer Science and Engineering
- M.Tech. in Computer Engineering from VJTI, Mumbai

□ Academic and Professional Experience

- Over 16 years of experience in academia and the engineering education domain
- Mentored numerous students at undergraduate and postgraduate levels

□ Areas of Specialization

- Operating Systems , User Experience Design (UxD), Data Science
- Soft Computing , Machine Learning, Artificial Intelligence
- Full Stack Development (MERN, Django, Flask)

□ Certifications and Recognitions

- **Elite NPTEL Online Certification (2019)** for academic excellence
- Delivered **expert sessions** in AICTE-ISTE approved FDPs and STTPs
- Served as an **evaluator** in hackathon

□ Research Contributions

- Authored 15+ research papers in Scopus-indexed international journals and conferences
- Reviewer for Springer and IEEE journals/conferences

□ Academic Roles and Memberships

- Open Elective Coordinator, Faculty Documentation etc.
- Active member of the Indian Society for Technical Education (ISTE)

Scheme

Course Code	Course Title							
116h54C701	Advanced Machine Learning							
	TH		P	TUT			Total	
Teaching Scheme(Hrs.)	03		2	--			05 ⁰³	
Credits Assigned	03		1	--			04 ⁰³	
Examination Scheme	Marks							
	CA		ESE	TW	O	P	P&O	Total
	ISE	IA						
	30	20	50	--	--	--	--	100

Outcomes

Course Outcomes

On completion of the course students will be expected to

1. Describe the fundamental issues and challenges of machine learning: data, model selection, model complexity.
2. Understand & describe mathematical foundation behind deep learning architectures.
3. Design & implement various deep supervised learning architectures for image data.
4. Learn and train various deep learning models in NLP.
5. Apply various deep learning techniques to design efficient algorithms for real-world applications.

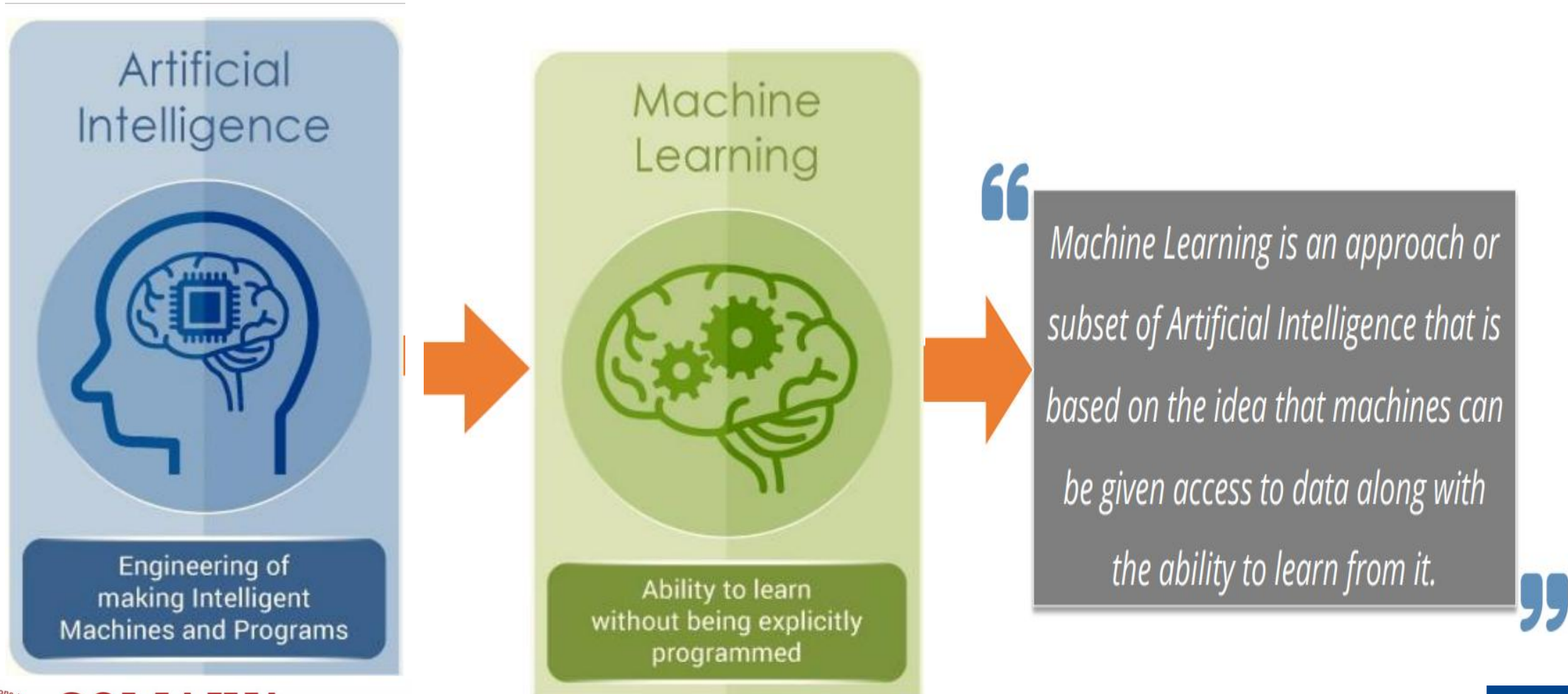
Syllabus

Module No.	Unit No.	Details	Hrs.	CO
1	Machine learning foundation:		05	CO1
	What is Machine Learning? Types of learning, applications, Bias, variance, overfitting, underfitting, cross validation and feature engineering, gradient descent learning algorithm and its variations			
2	Deep learning fundamentals:		10	CO2
	Mathematical foundations of deep learning & neural network playground. Learning rates and data normalization, activation functions, Optimizers, Regularization, Dropout, Momentum & Batch Norm. Neural Nets-Deep Vs Shallow Networks- deep multi-layer perceptron. #Self-Learning: forward and backward propagation in neural networks.			
3	Deep Learning Architecture:		10	CO3
	Convolutional Neural Networks: Convolution, pooling operations, basic CNN architecture and image classification using CNN. Recurrent Neural Networks: RNN introduction, difference between feed forward and RNN, forward, backward propagation in RNN & Vanishing/Exploding gradient problem. Long Short Term Memory: LSTM gates, forward, backward propagation in LSTM. #Self-Learning: Auto encoders			

Syllabus

4	Deep learning for NLP:	10	CO4
	NLP introduction, text preprocessing techniques, text representation (one hot encoding, bag of N grams, TFIDF features). Word Embedding – word2vec model. Text generation, Seq2Seq models (Encoder decoder architecture). <i>#Self-Learning: NLTK & Gensim library</i>		
5	Machine learning case studies:	10	CO5
	• Churn Analysis and Prediction • Sentiment Analysis or Topic Mining from New York Times • Customer Segmentation and Value • Netflix Movie Recommendation System • Image & text classification using TensorFlow & PyTorch. • Transfer learning with Pre-trained CNN using TensorFlow.		
Total		45	

Relationship between Artificial Intelligence and Machine Learning



What Is Machine Learning?

Machine Learning is the science (and art) of programming computers so they can learn from data.

[Machine Learning is the] field of study that gives computers the ability to learn without being explicitly programmed.

Arthur Samuel, 1959

What is machine learning? (continued)

And a more engineering-oriented one:

A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

Tom Mitchell, 1997

What is machine learning? (continued)

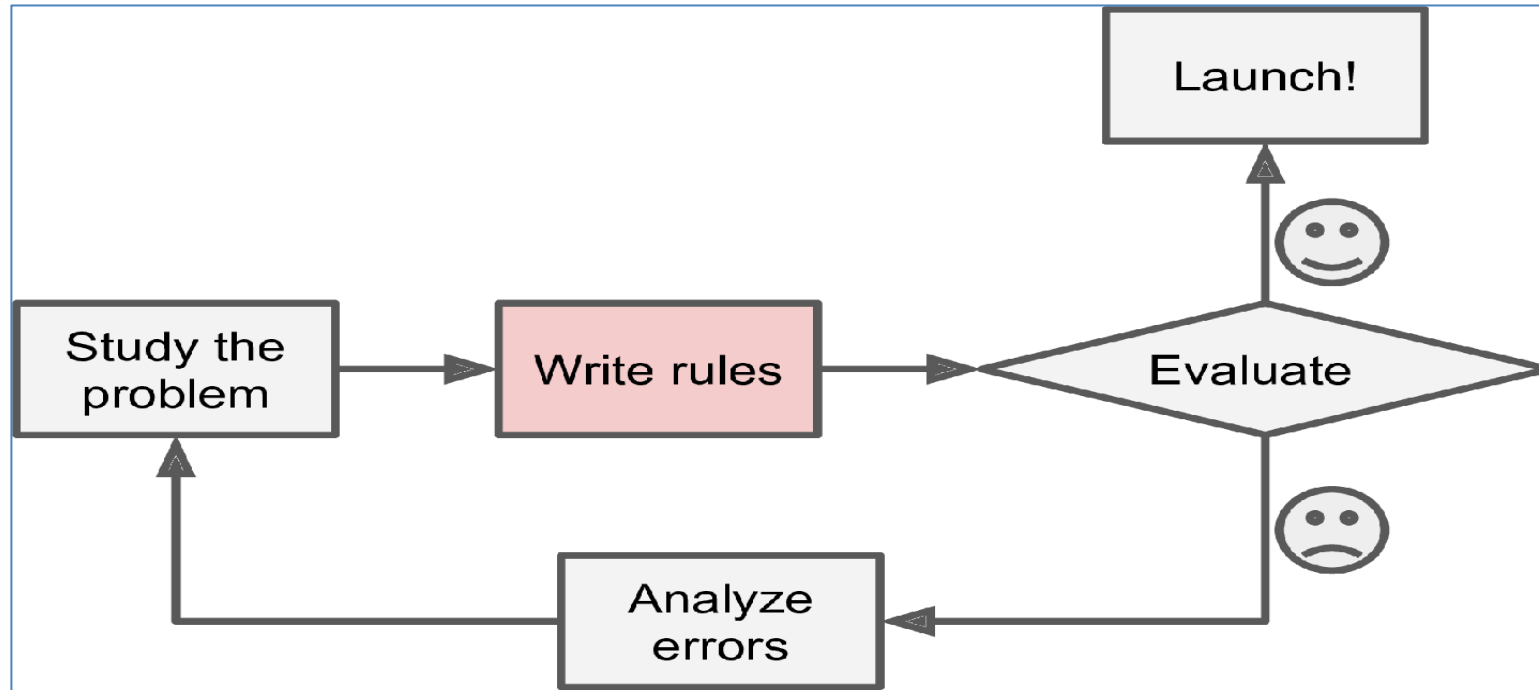
Machine Learning is the science of getting computer to learn and act like humans do, and improve their learning over time in autonomous fashion, by feeding them data and information in the form of observations and real-world interactions.



Example

- A **spam filter** is a Machine Learning program trained to **identify spam emails**.
- It uses examples of:
 - **Spam emails** (flagged by users)
 - **Non-spam emails** (also called **ham**)
- These examples form the **training set**, and each individual email in this set is called a **training instance** or **sample**.
- **Key Elements:**
 - **Task (T):** Flagging new emails as spam or non-spam
 - **Experience (E):** Learning from the training data (labelled emails)
 - **Performance Measure (P):** Evaluated using **accuracy**, calculated as the **ratio of correctly classified emails**.

Traditional Approach



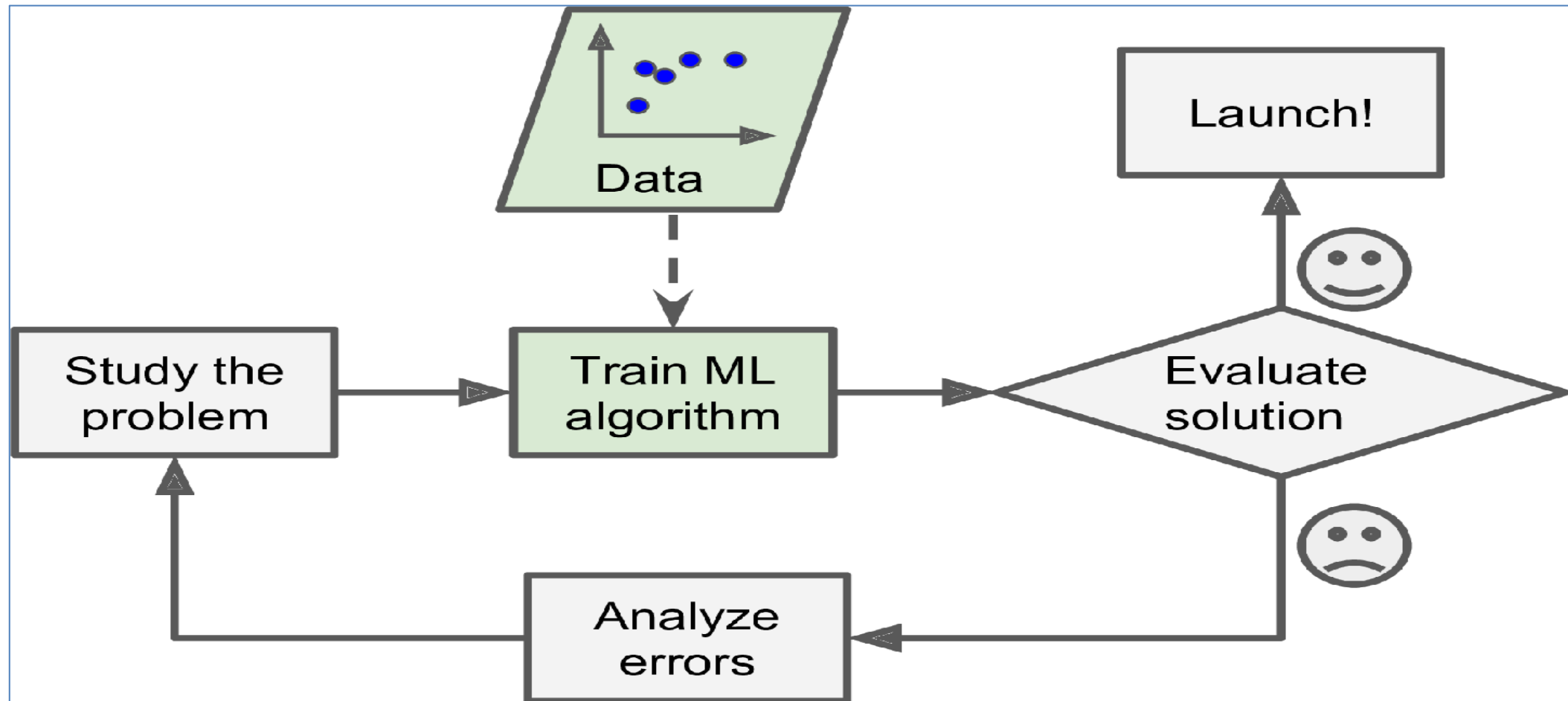
Traditional Programming



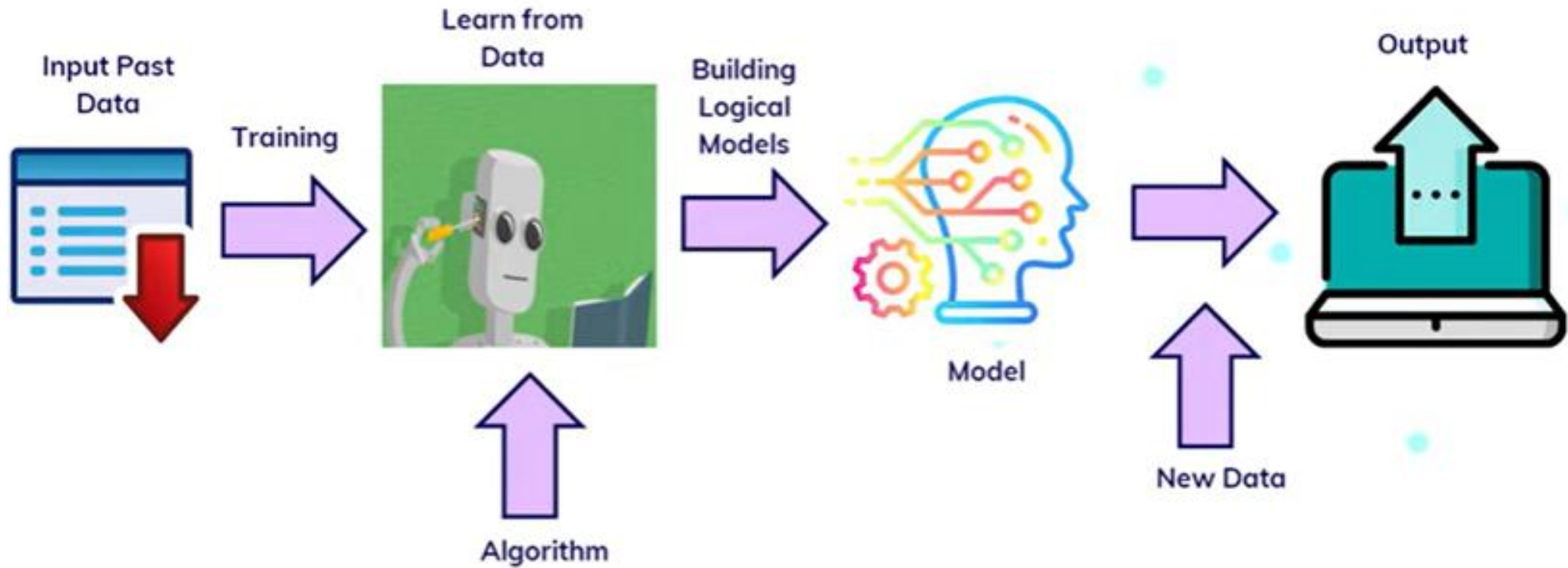
Tradition Way

- Starts with human experts **studying spam patterns**.
- Common identifiers:
 - Subject line: “Free”, “Win now”, “4U”
 - Sender patterns: Random characters or spoofed names
 - Body text: Excessive links, urgency, rewards
- Write rules like: IF "4U" IN subject AND has >2 links → flag as spam
- Use simple **if-then** logic
- Limitations
 - Difficult to scale and hard to maintain
 - Fragile to unseen variations

Machine Learning Approach



Machine Learning Approach (continued)



Machine Learning Approach

- **Uses labelled data:** thousands of emails tagged spam or ham (not spam).
- **Extracts features** like:
 - Word frequencies
 - HTML tags
 - Length of message
 - Sender domain, etc.
- **Trains algorithms** like:
 - Naive Bayes (text classification)
 - Logistic Regression
 - Support Vector Machines (SVMs)
- **Learns** which patterns correlate with spam.

Machine Learning

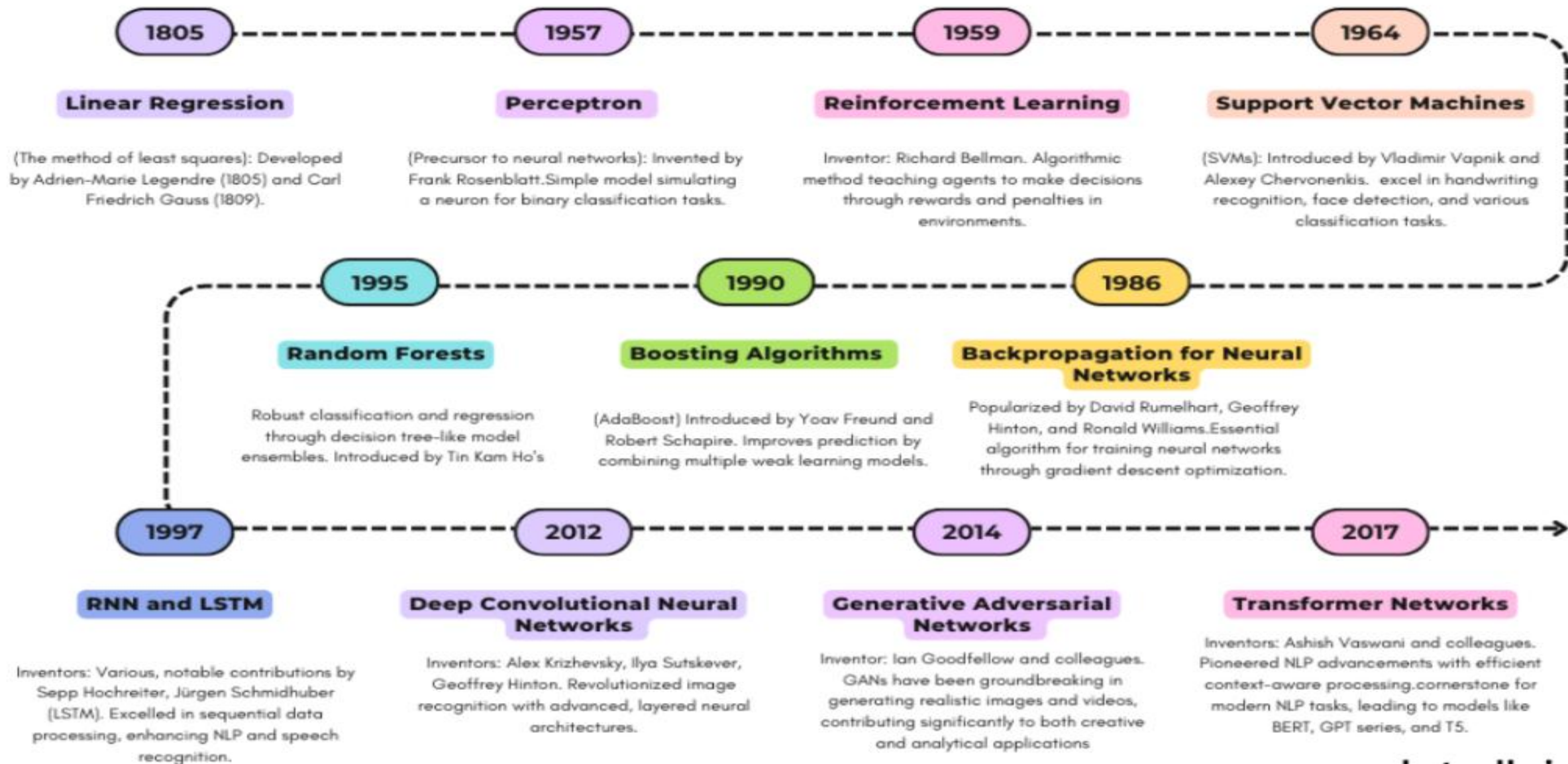
Advantages

- Resolving complex problems
- Automation for everything
- Trends and Pattern Identification
- Wide range of application

Disadvantages

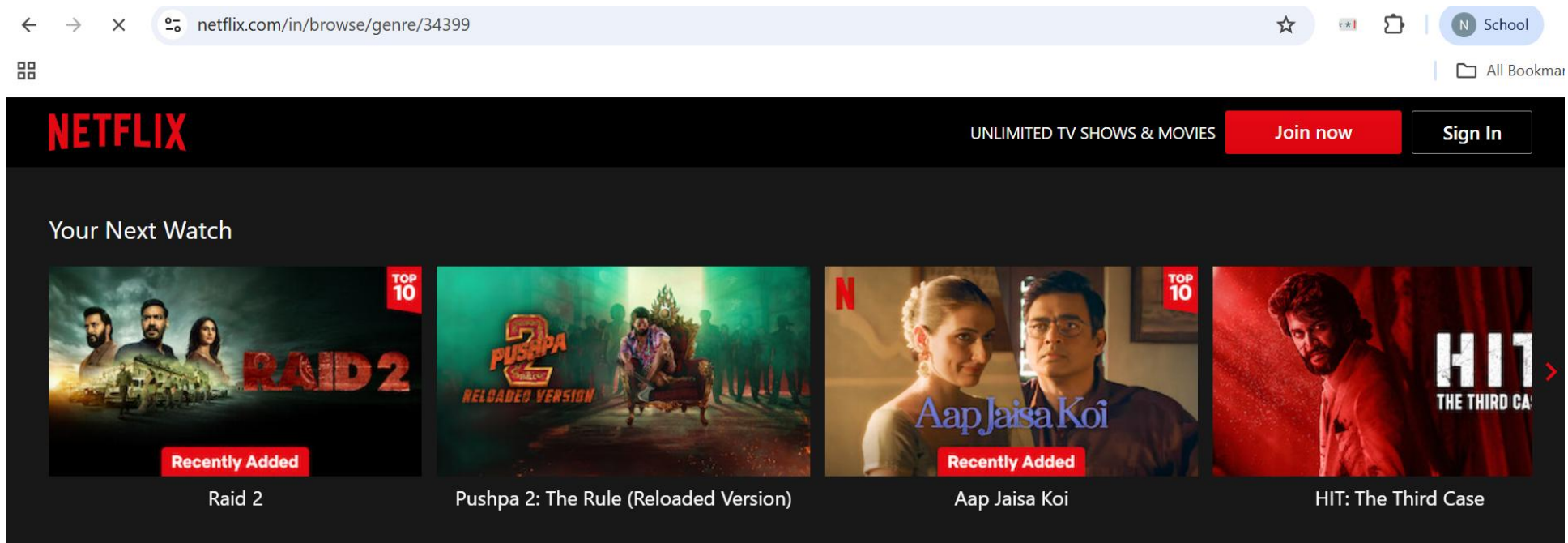
- Data Acquisition
- Highly Error-Prone
- Algorithm Selection
- Time - Consuming

Evolution of Machine Learning



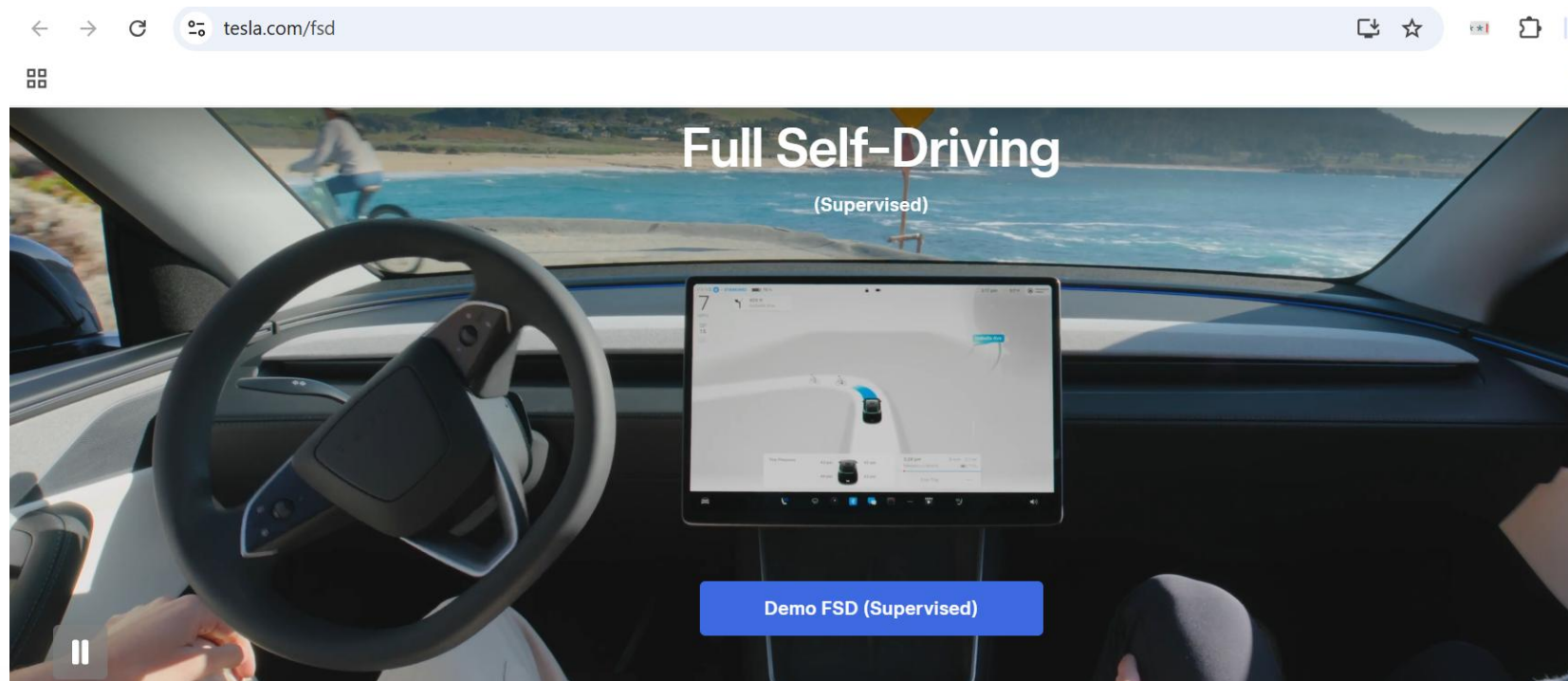
Recommendation Engines

- **Example:** Netflix's content recommendations
- **Application Area:** Media, Entertainment, Online Shopping



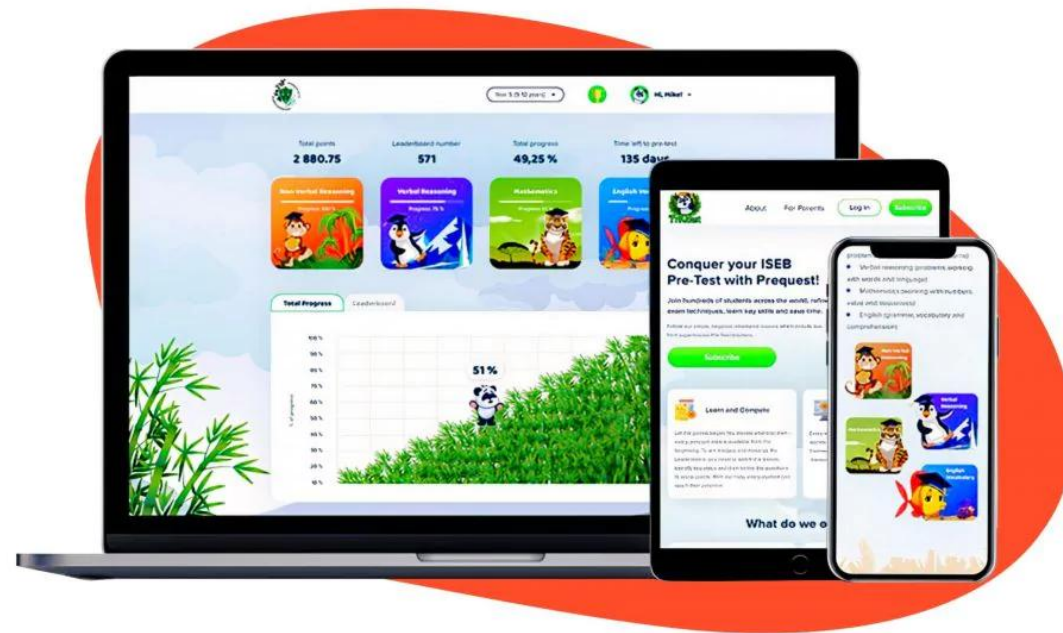
Self-Driving Cars

- **Example:** Tesla Autopilot
- **Application Area:** Automotive, Smart Mobility, Transportation



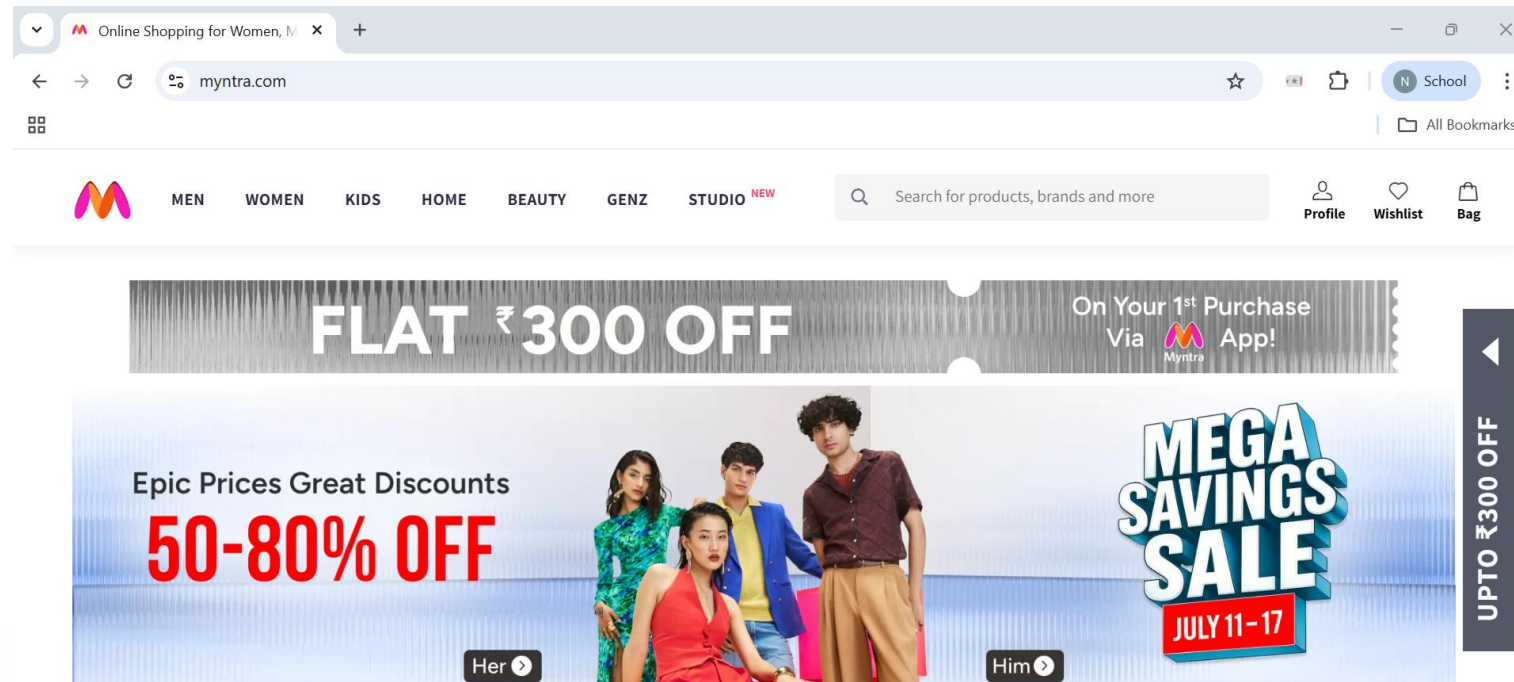
Gamified Learning and Education

- **Example:** Duolingo's Language Learning App
- **Application Area:** EdTech, Language Learning, Mobile Learning Platforms



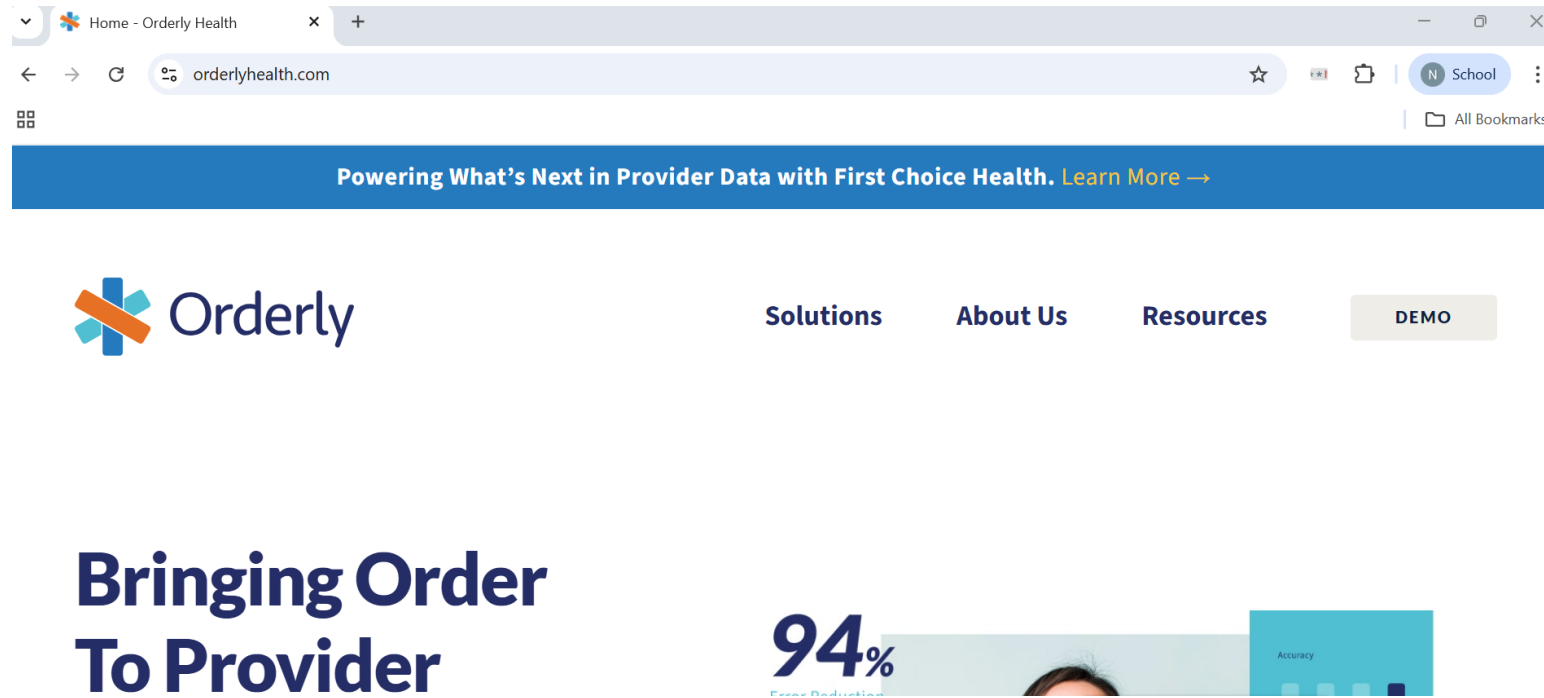
E-commerce Personalization

- **Example:** Myntra Product Recommendations and Search Personalization
- **Application Area:** Fashion E-Commerce, Online Retail, Personalized Shopping



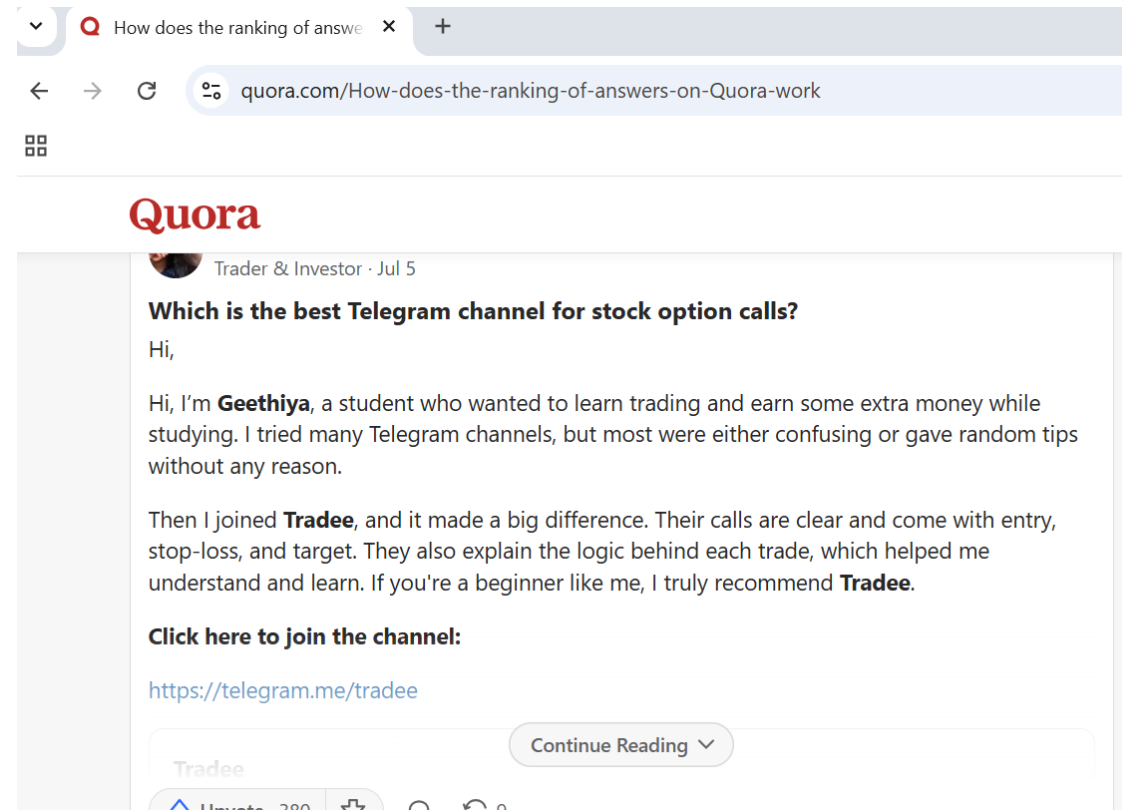
. Medical Diagnosis and Predictive Healthcare

- **Example:** Orderly Health (AI-driven medical query resolution)
- **Application Area:** Healthcare, Digital Health, Diagnostics



Smart Search and Answer Ranking

- **Example:** Quora's Super-Specific Answer Ranking System
- **Application Area:** Search Engines, Knowledge Platforms, Q&A Communities



Career Opportunities

- Machine Learning Engineer
- Business Intelligence Developer
- Data Scientist
- NLP Engineer
- Computer Vision

Question ?